RAJALAKSHMI ENGINEERING COLLEGE

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M2

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|---|---|
| **Started on** | Tuesday, 22 September 2026, 8:54 AM |
| **State** | Finished |
| **Completed on** | Tuesday, 29 September 2026, 8:37 AM |
| **Time taken** | 6 days 23 hours |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 10.00 out of 10.00

**Playing with Chessboard:**

Ram is given with an n*n chessboard with each cell with a monetary value. Ram stands at the (0,0), that the position of the top left white rook. He is been given a task to reach the bottom right black rook position (n-1, n-1) constrained that he needs to reach the position by traveling the maximum monetary path under the condition that he can only travel one step right or one step down the board. Help ram to achieve it by providing an efficient DP algorithm.

**Example:**

**Input**

3

**1** 2 4

**2** 3 4

**8 7 1**

**Output:**

19

**Explanation:**

Totally there will be 6 paths among that the optimal is

Optimal path value:1+2+8+7+1=19

**Input Format**

First Line contains the integer n

The next n lines contain the n*n chessboard values

**Output Format**

Print Maximum monetary value of the path

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int a[n][n];
    int dp[n][n];

    // Input
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            scanf("%d", &a[i][j]);
        }
    }

    dp[0][0] = a[0][0];

    // First row
    for (int j = 1; j < n; j++) {
        dp[0][j] = dp[0][j - 1] + a[0][j];
    }

    // First column
    for (int i = 1; i < n; i++) {
        dp[i][0] = dp[i - 1][0] + a[i][0];
    }

    // Fill DP table
    for (int i = 1; i < n; i++) {
        for (int j = 1; j < n; j++) {
            if (dp[i - 1][j] > dp[i][j - 1])
                dp[i][j] = dp[i - 1][j] + a[i][j];
            else
                dp[i][j] = dp[i][j - 1] + a[i][j];
        }
```

```
    }

    printf("%d", dp[n - 1][n - 1]);

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 3<br>1 2 4<br>2 3 4<br>8 7 1 | 19 | 19 | ✔ |
| ✔ | 3<br>1 3 1<br>1 5 1<br>4 2 1 | 12 | 12 | ✔ |
| ✔ | 4<br>1 1 3 4<br>1 5 7 8<br>2 3 4 6<br>1 6 9 0 | 28 | 28 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

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